As an example, the phase space propagation of Shardlow's method [333], may be written as

$$\exp\left(h\hat{\mathcal{L}}_{\text{DPD-S}}\right) = \exp\left(h\mathcal{L}_{\text{O}}\right)\exp\left(\frac{h}{2}\mathcal{L}_{\text{B}}\right)\exp\left(h\mathcal{L}_{\text{A}}\right)\exp\left(\frac{h}{2}\mathcal{L}_{\text{B}}\right), \tag{8.99}$$

where $h$ is the stepsize and, as usual, $\exp\left(h\mathcal{L}_f\right)$ represents the phase space propagator associated to the corresponding vector field $f$. In practice, Shardlow's method is typically implemented by further reducing the vector field O into interacting pairs:

$$\exp\left(h\hat{\mathcal{L}}_{\text{O}}\right) = \exp\left(h\mathcal{L}_{\text{O}_{\text{N}-1,\text{N}}}\right)\cdots\exp\left(h\mathcal{L}_{\text{O}_{1,3}}\right)\exp\left(h\mathcal{L}_{\text{O}_{1,2}}\right),$$

with the operators associated to each interacting pair defined as

$$\mathcal{L}_{\text{O}_{i,j}} = \left(-\gamma\omega^{\text{D}}(r_{ij})(\hat{\boldsymbol{q}}_{ij}\cdot\boldsymbol{v}_{ij}) + \frac{\sigma^2}{2}[\omega^{\text{R}}(r_{ij})]^2\hat{\boldsymbol{q}}_{ij}\cdot\left[\nabla_{\boldsymbol{p}_i} - \nabla_{\boldsymbol{p}_j}\right]\right)\hat{\boldsymbol{q}}_{ij}\cdot\nabla_{\boldsymbol{p}_i}. \tag{8.100}$$

Each interacting pair thus preserves the invariant distribution $\hat{\rho}_\beta$.

#### 8.9.2.3 DPD Trotter Scheme

In the stochastic DPD-Trotter scheme, the standard DPD system (8.90) is split into two parts, which are labelled "A" and "S" as indicated below

$$\mathrm{d}\begin{bmatrix}\boldsymbol{q}_i\\ \boldsymbol{p}_i\end{bmatrix} = \underbrace{\begin{bmatrix}m_i^{-1}\boldsymbol{p}_i\\ \boldsymbol{0}\end{bmatrix}\mathrm{d}t}_{\text{A}} + \underbrace{\begin{bmatrix}\boldsymbol{0}\\ \boldsymbol{F}_i^{\text{C}}\mathrm{d}t + \boldsymbol{F}_i^{\text{D}}\mathrm{d}t + \sigma\boldsymbol{R}_i\end{bmatrix}}_{\text{S}}. \tag{8.101}$$

The corresponding operator of part A is exactly the same as in Shardlow's method, while the operator of part S is actually the sum of the operators of B and O defined above

$$\mathcal{L}_{\text{S}} = \mathcal{L}_{\text{B}} + \mathcal{L}_{\text{O}}. \tag{8.102}$$

As in Shardlow's method, the vector field S is typically split into exactly solvable interacting pairs. Subtracting $\mathrm{d}\boldsymbol{v}_j$ from $\mathrm{d}\boldsymbol{v}_i$ and multiplying $\hat{\boldsymbol{q}}_{ij}$ on both sides gives

$$m_{ij}\mathrm{d}v_{ij} = F_{ij}^{\text{C}}(r_{ij})\mathrm{d}t - \gamma\omega^{\text{D}}(r_{ij})v_{ij}\mathrm{d}t + \sigma\omega^{\text{R}}(r_{ij})\mathrm{d}\text{W}_{ij}, \tag{8.103}$$